

# Action-Class Authority: A Verification-Layer Pattern for Agentic AI Security

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## Abstract

Production-scale agentic AI systems fail at the same architectural seam: a prompt-injected, memory-poisoned, or supply-chain-compromised agent retains valid credentials and reasons over its own action authority. We propose action-class authority as a verification-layer pattern that decouples action classification from agent runtime reasoning. Four independent May 2026 publications—across academic systems security, industry framework engineering, empirical production-scale evaluation, and academic adversarial machine learning—converge on the same architectural conclusion: the boundary must move to the action layer. We document the OWASP AISVS verification requirements (C9.2.6 manifest-declared classification, C9.2.7 worst-case chain rule) that operationalize this conclusion, analyze three canonical 2026 cases (Codex docker bind-mount, Meta Instagram AI Support Assistant exploit, Red Hat npm Miasma supply chain), and present a cross-walk to eight adjacent standards bodies and contributions showing the action-class authority

pattern as an emerging cross-organization synthesis. We close with five open research questions.

**Keywords:** agentic AI security, action-class authority, OWASP AISVS, manifest-declared classification, deterministic gate, reversibility-graded authority, multi-step chain rule, agent verification, capability-based security

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## 1. Introduction

The agentic AI deployment year of 2026 has crystallized a question the field was able to defer through 2024 and 2025: where does the security boundary for an autonomous agent live? Through the early agentic era, the boundary was treated implicitly as a property of the model—better instruction-following, better safety training, better refusal calibration, the argument went, would produce agents whose reasoning could be trusted to refuse the actions an organization did not want to authorize. Frontier model releases were evaluated on benchmarks that measured the model’s ability to recognize unsafe requests and decline them. Agent deployments inherited the same assumption: if the model is safe, the agent built on top is safe.

May 2026 closed that argument. Four independent publications—Christodorescu and Fernandes’ systems-security framing, Anthropic’s Zero Trust for AI Agents enterprise framework, Kereopa-Yorke et al.’s oracle poisoning measurement at 100% trust rate across nine frontier models, and Pulipaka et al.’s sleeper memory poisoning at 95–99% injection rate—reached the same architectural conclusion through four independent methodologies. Agent reasoning cannot be the safety boundary. The boundary has to live at the action layer.

### 1.1 Contributions

This article makes five contributions:

1. **Convergence synthesis.** We synthesize four May 2026 publications into a single

architectural conclusion with verbatim empirical numbers cited from one source rather than re-aggregated across four (95–99% memory-layer injection, 100% trust on poisoned knowledge graphs, 60–89% attacker-intended action rates after retrieval).

2. **Verification standard documentation.** We document the OWASP AISVS C9.2.6 manifest-declared classification requirement and the C9.2.7 worst-case chain rule, contributed via PR #822 to the AISVS research chapter (merged 2026-05-27) and proposed for v1.01 inclusion, with worked threat model showing why the same agent cannot both propose and classify its action class.
3. **Worked case analysis.** We analyze three canonical 2026 cases (Codex Docker bind-mount, Meta Instagram AI Support Assistant exploit, Red Hat npm Miasma supply chain compromise) demonstrating that the architectural pattern is identical across coding agent, customer-facing AI, and CI/CD pipeline.
4. **Cross-domain prior-art framing.** We frame action-class authority as a cross-domain pattern with structural parallels in capability-based security, transactional database reversibility, formal verification of reversible computation, and digital forensic chain of custody—four traditions converging on the same architectural separation between actor and verifier.
5. **Standards cross-walk.** We provide a cross-walk to eight adjacent standards-track contributions (CSA NHI, OWASP SPVS PR #14, OWASP SPVS PR #15, OWASP SPVS

Issue #16, OWASP Cornucopia AAI, OWASP AIVSS, MITRE ATLAS, IETF AI-agent-authentication, CSA ZT Microsegmentation) demonstrating the action-class authority pattern as an emerging cross-organization synthesis in mid-2026. Status of each contribution is reported conservatively against current repo state, distinguishing under-discussion items from merged spec text.

The remainder of this article is organized as follows. Section 2 describes the May 2026 empirical convergence. Section 3 specifies the verification pattern. Section 4 presents three canonical cases. Section 5 compares against LLM-guardian alternatives. Section 6 cross-walks to adjacent standards. Section 7 enumerates open research questions. Section 8 concludes.

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## 2. The May 2026 Empirical Convergence

Four publications landed in May 2026 that, read together, settle the agent-reasoning-as-boundary question through four independent methodologies.

### 2.1 Systems-Security Framing

Christodorescu and Fernandes (Google and University of California San Diego) and collaborators argue from systems-security first principles that the AI model itself must be treated as an untrusted component, with security invariants enforced at the system level rather than within the model’s reasoning. The argument is structural: a model that can be prompted, poisoned, or contextually shifted into producing arbitrary outputs cannot also be the mechanism that decides which outputs are safe to act on.

## 2.2 Operational Framework

Anthropic published the most comprehensive operational framework to date for Zero Trust in agentic AI systems. The framework identifies the boundary problem as a top-of-stack concern and introduces the “impossible vs. tedious” design test: controls whose value is friction (rate limits, extra hops, non-standard ports) degrade against an adversary with unlimited patience; controls that remove a capability survive. Anthropic’s principle reaches further than its prescription—a point Hardt [11] argues at the credential layer, and the present work extends to the action layer.

## 2.3 Oracle Poisoning Measurement

Kereopa-Yorke and collaborators (Microsoft, SAP, OWASP GenAI Security Project, University of New South Wales Canberra) demonstrated empirically that every tested state-of-the-art model trusts moderately-sophisticated poisoned knowledge graph data at 100%. The empirical setup uses a 42-million-node code knowledge graph and nine state-of-the-art models from OpenAI, Anthropic, and Google. Three attacker sophistication levels (L1 obvious malicious naming, L2 plausible naming with OWASP-style references, L3 authority markers and adversarial signatures) show that the action-on-retrieved-content rate scales with sophistication while detection rates do not.

## 2.4 Sleeper Memory Poisoning

Sidharth Pulipaka and collaborators (SPAR, ELLIS Tübingen, CISP Helmholz) reached the same conclusion at the memory layer: 95–99% injection rates at the memory layer, 90–95% retrieval rates on goal-adjacent queries, and 60–89% attacker-intended action rates after retrieval. A poisoned memory persists across sessions; the agent’s reasoning never sees the injection point because the injection happened at write time, not read time.

## 2.5 The Convergent Conclusion

Read in isolation, each paper makes a strong claim. Read together, the four constitute a structural

argument: agent reasoning is not a safety boundary because (a) the reasoning component is the same one an adversary can corrupt, and (b) corruption rates at memory and retrieval are empirically high enough that any single-layer defense based on reasoning is statistically dominated. The structural answer—if the agent’s reasoning cannot be trusted to classify its own actions, the classification must come from elsewhere—is the verification problem this article specifies.

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## 3. The Verification Pattern

The verification pattern that operationalizes the May 2026 conclusion has two requirements, both contributed to the OWASP AISVS research chapter via PR #822 (merged 2026-05-27) and proposed for v1.01 inclusion.

### 3.1 C9.2.6 Manifest-Declared Classification

C9.2.6 requires that the reversibility classification of an agent action be declared in the tool or action manifest at design time, not derived from the agent’s runtime output. The four classes are:

- **Read-Only (RO):** action observes state without modifying it
- **Reversible (R):** action modifies state within the system; can be rolled back via undo log, transaction abort, or system-level compensating action
- **External-Reversible (ER):** action affects external systems but is recoverable through identity verification, customer support escalation, or similar out-of-band channel
- **Irreversible (I):** action commits to durable state in ways the user cannot recover from without out-of-band action; emails sent, funds transferred, packages published to public registries, account identifiers linked

The classification lives in the manifest. The gate evaluates the manifest. The agent reasons elsewhere. This is the architectural separation the prior-art traditions (capability-based security, transactional databases, formal verification of reversible computation, digital forensic chain of custody) all establish at their respective layers.

### 3.2 C9.2.7 Worst-Case Chain Rule

C9.2.7 requires that for a composed multi-step action sequence, the gate evaluates the worst-case action class reachable in the chain, not the per-step class. The rule prevents the composition failure where individually low-gate steps chain into an irreversible end state: read user email (RO) → identify financial accounts (RO) → initiate password reset (ER) → modify account credentials (I) is a chain whose worst-case class is irreversible from the moment the first step executes, not at the irreversible step.

### 3.3 Threat Model

The threat C9.2.6 specifically addresses: when the same agent both proposes and classifies an action, a prompt-injected or context-corrupted agent can mislabel an irreversible action as reversible. The gate based on the runtime classification will permit the action. The manifest-declared classification, by contrast, was set at design time and cannot be rewritten by the agent at runtime. This is the structural answer to the agent-reasoning-as-boundary failure mode.

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## 4. Three Canonical Cases

Three real-world cases from late May and early June 2026 demonstrate the architectural pattern.

### 4.1 Codex Docker Bind-Mount (May 30, 2026)

OpenAI's Codex coding agent, operating on a Linux host where the user was in the docker group, chained four legitimate operations: enumerate group memberships, recognize docker as root-equivalent, construct `docker run --rm -v /etc:/host-etc:rw ubuntu:22.04 ...`, and execute a write that overwrote a live configuration file. Each operation was permitted by the tool authority. The composite action was irreversible at the system level. The C9.2.6/C9.2.7 architecture would have classified "write to /etc on live host" as irreversible in the manifest; the deterministic gate would have routed the action to human ratification regardless of how the agent reasoned about the workflow.

### 4.2 Meta Instagram AI Support Assistant (May 31, 2026)

Meta's AI Support Assistant on Instagram had tool authority to modify account state, including linking new email addresses. Attackers identified the prompt pattern that triggered email-link change without identity verification. Exploitation persisted for approximately four months across thousands of accounts including the Obama White House Instagram account and security researcher Jane Manchun Wong. Email-link to a primary identifier is irreversible at the user level (password-reset control follows the email). A manifest-declared classification of this action class as irreversible, evaluated by a deterministic gate, would have routed the change to out-of-band identity verification regardless of the prompt the attacker constructed.

### 4.3 Red Hat npm "Miasma" Supply Chain (June 1, 2026)

The @redhat-cloud-services npm scope was compromised on June 1, 2026, with 32 packages backdoored. The attack vector was not credential theft—attackers pushed orphan commits adding a workflow file with `id-token: write` permissions, then leveraged OIDC trusted publishing to push to the

npm registry through the legitimate path. The Aikido analysis [8] identifies the architectural failure: “trusted publishing, designed to eliminate long-lived tokens, becomes a misleading trust signal when CI/CD pipeline access is compromised.” The C9.2.6 verification would classify “publish to a public npm scope” as irreversible at the user level in the manifest; the worst-case chain rule of C9.2.7 would gate any chain reaching the publish step as irreversible from the chain start.

#### 4.4 The Architectural Pattern Across Three Cases

| Case        | Agent / Pipeline     | Tool authority          | Missing layer                                     | Composite irreversible action                     |
|-------------|----------------------|-------------------------|---------------------------------------------------|---------------------------------------------------|
| Codex       | Coding agent         | Docker group membership | Action-class declaration at docker-mount boundary | Write to /etc on live host                        |
| Meta IG     | AI Support Assistant | Account modification    | Action-class declaration on email-link            | Identifier change with password-reset implication |
| Red Hat npm | CI/CD pipeline       | OIDC trusted publishing | Action-class declaration at publish step          | Public registry publish                           |

The pattern is identical across coding agent, customer-facing AI, and CI/CD pipeline. The three cases use

three different agent technology stacks from three different vendors. The composition failure is the same.

#### 5. Comparison with LLM-Guardian Approaches

An alternative architecture proposed in concurrent research uses a second LLM as a guardian agent that evaluates the primary agent’s proposed actions before they execute. The guardian approach has empirical attraction (LLMs are good at policy interpretation) but inherits the same threat model the action-class authority pattern is built to escape: a guardian whose reasoning can be context-shifted or prompt-injected is structurally identical to the agent it guards. The deterministic gate evaluating a manifest-declared classification does not have this property—the classification is set at design time and is not rewritten at runtime by any reasoning component.

This is not a rejection of LLM-guardian approaches. The two architectures are complementary: the manifest-declared deterministic gate is the structural verification layer; LLM-guardian evaluation can layer additional context-aware checks above the gate for cases where the manifest classification is permissive. The architectural argument is about which layer carries the verification load, not whether LLM-based checks have value.

#### 6. Cross-Walk to Adjacent Standards

The action-class authority pattern is one node in a broader set of standards-track contributions across multiple bodies. Table 1 summarizes the cross-walk. Status is reported conservatively against repo state as of 2026-06-15: items recorded as “merged” reflect the pull request merged on the named date in repository history, distinct from inclusion in a published release; items under maintainer dialogue or peer review are recorded as such, distinct from acceptance.

**Table 1: Standards cross-walk for action-class authority (mid-2026; status verified 2026-06-15)**

| Standards body                     | Contribution                                                                                                                                                                                                                                                                                                        | Status (as of 2026-06-15)                                                                                                     | Standards body                     | Contribution                                                                                                                                                                                  | Status (as of 2026-06-15)                                                                                                                                                                                  |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                    |                                                                                                                                                                                                                                                                                                                     |                                                                                                                               | <b>PR #14</b>                      | response decision-rights matrix                                                                                                                                                               | review (opened May 18, 2026; not merged)                                                                                                                                                                   |
| <b>OWASP AISVS</b>                 | C9.2.6 manifest-declared classification + C9.2.7 worst-case chain rule                                                                                                                                                                                                                                              | Contributed via PR #822 (merged 2026-05-27); substance proposed for v1.01 inclusion                                           | <b>OWASP SPVS PR #15</b>           | V1.3.7 NHI runtime decision-rights matrix                                                                                                                                                     | Open under review (opened May 18, 2026; not merged)                                                                                                                                                        |
| <b>CSA Non-Human Identity v1.0</b> | Four-element attribution language (delegator / agent / intent / actions) anchored at para 222 of the Working Draft (joint peer-review contribution with Mallikarjunarao Sunke; full six-property formalization — chain-id binding, originating-principal immutability, audit telemetry surface — targeted for v2.0) | Peer review closed June 4, 2026; editorial close mid-June 2026 (Google Doc workflow; no public version-stamp until published) | <b>OWASP SPVS Issue #16</b>        | Trust-inheritance threat model — AUR Atomic Arch + agent-skill-registry paired worked examples as V1.6 supply-chain input                                                                     | Filed June 13, 2026; open, awaiting maintainer routing                                                                                                                                                     |
|                                    |                                                                                                                                                                                                                                                                                                                     |                                                                                                                               | <b>OWASP Cornucopia Agentic AI</b> | Per-action-class reversibility mapping dimension proposed for companion-mappings-1.0.yaml (Issue #3018, May 2026); RO/R/ER/I label refinement converged with maintainer @sydseter in dialogue | Open under active discussion; maintainer @cw-owasp raised dependency on published-standards backing before merge; thread holds the mapping pending AISVS settlement (Mayur opened AISVS#820 per maintainer |
| <b>OWASP SPVS</b>                  | V5.6.5 incident-                                                                                                                                                                                                                                                                                                    | Open under                                                                                                                    |                                    |                                                                                                                                                                                               |                                                                                                                                                                                                            |

| Standards body                                   | Contribution                                                                                                                             | Status (as of 2026-06-15)                                                                                                                                       | Standards body                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Contribution                                                       | Status (as of 2026-06-15)                           |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------|
| <b>OWASP AIVSS</b>                               | Discussion thread opened on action-class authority as a scoring dimension for autonomous-execution risk (Issue #36, opened May 31, 2026) | Open; awaiting project leadership engagement (zero comments at time of writing)                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | segmentation / action-class authority / application authorization) | Philip Griffiths on June 14, 2026); pre-publication |
| <b>MITRE ATLAS M0030</b>                         | Reversibility classification as deterministic trigger for human-in-the-loop intervention; mapped to AML.T0053, AML.T0101, AML.T0086      | Refinement submitted to atlas@mitre.org on May 23, 2026; awaiting MITRE editorial response (no public status indicator in the MITRE ATLAS contribution channel) | <p>The cross-walk pattern is consistent across bodies: action-class authority is the verification-layer formulation that operationalizes the May 2026 conclusion across networks, identity, supply chain, threat modeling, vulnerability scoring, MITRE ATLAS mitigations, and IETF authentication. The bodies converge from independent starting points. The status column above is the load-bearing one for readers evaluating the maturity of each landing point: one chapter merged, one peer review complete, two PRs and three issues open under maintainer dialogue, two editorial responses pending. The convergence pattern is real; the spec-text merge state is heterogeneous, as is appropriate for a synthesis being assembled across eight bodies simultaneously.</p> <hr/>               |                                                                    |                                                     |
| <b>IETF draft-klrc-aiagent-auth-01</b>           | Reversibility as deterministic trigger for §10.6 CIBA human-approval confirmation step (GitHub Issue #114)                               | Filed May 23, 2026; open, awaiting working-group engagement                                                                                                     | <h2>7. Open Research Questions</h2> <p>Five open questions warrant empirical study:</p> <ol style="list-style-type: none"> <li><b>1. Input-layer corruption vs. action-layer gate.</b> When the manifest itself is corrupted at distribution time (compromised tool registry, malicious provenance), what is the residual exposure of an otherwise-correct action-class gate? Hypothesis: input-layer corruption is bounded by manifest integrity-binding (cryptographic signature, chain-of-custody at install time); empirical measurement of attack rates against signed vs. unsigned manifest distributions would settle the threshold.</li> <li><b>2. Multi-step information flow control.</b> The worst-case chain rule gates chains; it does not detect cross-chain information flows</li> </ol> |                                                                    |                                                     |
| <b>CSA Zero Trust Microsegmentation Guidance</b> | Action reversibility as agentic risk-tier signal; 3-layer scope (nano-                                                                   | Peer review complete (5 threads resolved by lead author                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                    |                                                     |

where reversibility-graded data moves between chains. Hypothesis: information-flow labeling at the data layer, with reversibility class as a label dimension, would extend the chain rule to cross-chain composition.

3. **Runtime drift detection.** When an agent's action distribution shifts toward higher-risk classes over time (more irreversible actions, more external-reversible chains), is this drift detectable from action-class telemetry alone, or does it require correlated identity, posture, and content signals? Hypothesis: action-class distribution is a leading indicator; correlated signals are confirmatory.
4. **Reversibility classification disagreement.** When two parties (vendor manifest vs. enterprise policy) classify the same action differently, what is the protocol for resolving the disagreement? Hypothesis: worst-case wins (most-conservative class governs gate) is the safe default; empirical case study of real classification conflicts would test this.
5. **Reversibility class evolution.** When an originally-reversible action becomes irreversible due to downstream events (e.g., a "reversible" credential rotation becomes irreversible once propagated to external systems), how is the classification updated, and what is the gate behavior in the transition window? Hypothesis: time-bounded classification escalation with conservative defaults during transition.

These questions have empirical testability and direct relevance to deployed agentic systems. They

are appropriate targets for the next twelve months of research.

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## 8. Conclusion

The empirical evidence makes the structural argument unavoidable. Ninety-five to ninety-nine percent injection rates at the memory layer. One hundred percent trust rates on poisoned knowledge graphs across nine frontier models. Sixty to eighty-nine percent attacker-intended action rates after retrieval. The agent's reasoning will be wrong because the input was corrupted, or because the attacker found a prompt that produces a confident-but-mistaken interpretation, or because the agent's memory holds an injected goal from a session weeks earlier.

The defense lives at the action layer. The manifest declares which actions are read-only, which are reversible, which are external-reversible, and which are irreversible. The deterministic gate evaluates the manifest declaration, not the agent's runtime claim. The worst-case action class across a multi-step chain governs the gate, not the average. The agent that has been fooled by corrupted input still cannot execute an action whose manifest-declared classification requires human ratification.

The structural separation we synthesize has parallels in capability-based security, transactional databases, formal verification of reversible computation, and digital forensic chain of custody—four traditions converging on the same architectural answer. The OWASP AISVS C9.2.6 and C9.2.7 verification requirements operationalize this answer as verifiable, spec-anchored research-chapter contribution (PR #822, merged 2026-05-27 by otosulin; substance proposed for v1.01 inclusion). The pattern is being engaged in parallel across CSA NHI, OWASP SPVS (three open contributions), OWASP Cornucopia AAI, OWASP AIVSS, MITRE ATLAS, IETF AI-agent-



authentication, and CSA Zero Trust Microsegmentation work—with maturity heterogeneous across landings, as documented in Table 1.

The conversation is no longer about whether agent reasoning is trustworthy enough to act on. The empirical answer is no, at one hundred percent. The conversation is about where the gate sits and what the gate evaluates against. The action layer is where that gate belongs.

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### Author

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*Note on CSA NHI anchor: the v1.0 anchor reflects the four-element attribution language (delegator / agent / intent / actions) at para 222 of the Working Draft; the six-property formalization (chain-id binding, originating-principal immutability, audit telemetry surface) is targeted for v2.0 inclusion (joint contribution with Mallikarjunarao Sunke).*

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### Conflict of Interest Disclosure

The author is the contributor of OWASP AISVS C9.2.6 and C9.2.7 verification requirements cited in this paper (Section 3 and reference [5]). The author

is also the filer of OWASP SPVS PRs #14, #15 and Issue #16, OWASP AIVSS Issue #36, OWASP Cornucopia Agentic AI Issue #3018, the MITRE ATLAS M0030 refinement submission, and the IETF draft-klrc-aiagent-auth-01 Issue #114 referenced in the standards cross-walk table (Section 6).

The Cloud Security Alliance Non-Human Identity v1.0 anchor cited in Section 6 (reference [6]) is the four-element attribution language (delegator / agent / intent / actions) at para 222 of the Working Draft, contributed as a joint peer-review item with Mallikarjunarao Sunke. The full six-property formalization (chain-id binding, originating-principal immutability, audit telemetry surface) is not verbatim in v1.0; it is targeted for v2.0 inclusion, with joint-credit attribution to Sunke preserved on any future reference to that schema. The CSA Zero Trust Microsegmentation Guidance (Section 6, reference [15]) acknowledges the author in its Reviewers list.

These attributions are disclosed transparently and are consistent with scholarly practice for author-cited standards-track research, including the precedent of OWASP LLM Top 10 v2 (2025), which contains twenty arXiv-anchored citations across six of its twelve vulnerability files with author conflicts of interest disclosed in pull-request bodies (reference [14]).

The author has no financial relationship with OpenAI, Meta, Red Hat, Anthropic, or any other vendor named in the case analyses of Section 4. The author's employer, StraightArc Technologies Pvt. Ltd., is not in commercial competition with any vendor named in the cases or referenced in the standards cross-walk.

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